“Nav” Navaneethakannan Arumugam

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SUMMARY

Data and AI engineer who builds production systems end to end, from cloud data platforms to applied AI. Designs medallion and star-schema data models, orchestrates batch pipelines with Apache Airflow (Cloud Composer) on Google Cloud and Snowflake and event-driven Extract-Transform-Load (ETL), and keeps data trustworthy through Monte Carlo observability, automated testing, and Continuous Integration/Continuous Deployment (CI/CD). Builds Generative AI (GenAI) applications on Large Language Models (LLMs), from transformer-based Natural Language Processing (NLP) to Retrieval-Augmented Generation (RAG) agents on Databricks and open-source stacks that are evaluated, traced, grounded in cited sources, and connected to enterprise data through the Model Context Protocol (MCP), and turns manual workflows into automated, monitored platforms that teams rely on as their single source of truth.

EXPERIENCE

Analyst (Data and AI Engineering), Blackbuck Insights (BBI) Sep 2026 – Present

Data and AI engineering for client delivery; IL, USA

- Built a bronze/silver medallion e-commerce data platform on Google Cloud (Cloud Storage, Cloud Composer) and Snowflake covering 10 source entities and 23 tables and views, by designing the data contract, ingestion flow, and bronze-to-silver layers from scratch.
- Automated daily ingestion of 4,500–4,800 records per run (plus a 56,963-record go-live snapshot; 70,860 records loaded to date) with full load traceability by building two Apache Airflow DAGs: one generates synthetic batches into Cloud Storage, the other validates each batch with 18 automated checks and loads Snowflake bronze tables with per-table audit logging (144 audited loads), covered by unit tests.
- Optimized daily ingestion to run end to end in about 1–1.5 minutes (all 10 tables in 39–46 seconds) by redesigning the Airflow DAG into a single-session Snowflake load task with timeouts and fail-fast handling of non-retryable errors.
- Delivered a curated (silver) layer that reconciles 100% of 68,166 distinct bronze keys with zero rejected rows, excluding 2,694 superseded record versions (3.8%), by building 11 Snowflake Dynamic Tables with key validation, latest-version deduplication, 45-column type standardization, and lineage tagging, each refreshing automatically in 0.6–3.0 seconds.
- Established end-to-end data observability with 88 Monte Carlo monitors (freshness, volume, schema, and field quality) across 23 tables in 3 schemas, by integrating the bronze, silver, and audit layers and both Airflow DAGs into Monte Carlo for data-quality and pipeline-health monitoring.
- Built a fully local, zero-cost RAG agent over a 67-page technical PDF (223 heading-aware chunks) that achieved a 100% retrieval hit rate@4 and zero false refusals on a 15-question evaluation set, using hybrid ChromaDB and BM25 retrieval fused with reciprocal rank fusion and Ollama-served qwen3:8b with nomic-embed-text embeddings.
- Grounded every answer in page-level citations and enforced exact “Not in the document” refusals for out-of-scope questions, verified by an automated evaluation harness comparing dense and hybrid retrieval.
- Instrumented the agent with OpenTelemetry (a 6-span trace per question capturing retrieval, embeddings, prompts, tokens, and latency) and visualized traces in a self-hosted Arize Phoenix instance, using span timings to isolate LLM generation as the latency bottleneck.
- Connected AI clients to the Monte Carlo data-observability platform and the RAG agent through the Model Context Protocol (MCP), extending governed observability data and document knowledge to AI assistants on demand.

Software Developer, Merit Controls Jan 2026 – Aug 2026

Industrial control systems integration for renewable-energy assets; NJ, USA

- Established the single source of truth for vendor device data, adopted across the product team to validate SCADA, PVPC, and MPC releases, by independently architecting the OEM database: a star-schema PostgreSQL model normalizing 20,000+ Modbus/DNP3 registers from 15+ Battery Energy Storage System (BESS) and PV-inverter manufacturers into 1,500+ canonical tags via a transaction-safe, auto-triggering ETL.
- Designed and shipped the PVPC Transformer, a full-stack code-generation platform (Python, FastAPI REST API, React frontend) that auto-generates AcSELerator RTAC configuration files for 20+ devices, cutting Power Plant Controller (PPC)/SCADA commissioning from weeks of manual engineering to minutes; v1.0/v1.x generate client files from the OEM database and driver files from templates, with v2.0 (underway) generating both directly from the database.
- Kept that dataset current with zero manual intervention by orchestrating an event-driven pipeline in n8n (Docker Compose, Flask API) that triggered on new device-data ingestion, re-ran the three-stage ETL behind a caching layer that held refresh times down, regenerated downstream artifacts, and executed the test suite as one automated build-test-deploy cycle.
- Caught data and schema regressions before every release by building GitHub Actions CI/CD pipelines running 100+ automated unit, integration, and regression tests (pytest, vitest) across the ETL and the tools that consume it.
- Kept the platform trustworthy in production by handling upstream vendor schema drift, retrying transient ingestion failures, alerting on load errors, and supporting backfills that reprocess historical device data without corrupting canonical tags.
- Reduced requirements investigation and QA/QC review from hours of searching 100-page documents to page-cited answers by architecting a fully local RAG pipeline over unstructured PDFs (Tesseract OCR, all-MiniLM-L6-v2 embeddings, Chroma vector store, hybrid dense/BM25 retrieval), upgrading the generator from SmolLM-135M to Qwen 1.5B Instruct for answer quality within offline hardware limits, at zero per-query cost.

AI Engineer, Copani LLC Jan 2025 – Jan 2026

AI-powered agentic workflows for customer-service automation; FL, USA

- Automated triage of 50+ daily customer emails at roughly 90% accuracy by building a Python NLP pipeline on Hugging Face Transformers that encodes messages with BERT and maps them to a curated response library via semantic matching.
- Increased email-processing efficiency 90% by integrating the pipeline with the Gmail API into an end-to-end automated workflow that classifies intent and drafts client-specific responses under data-security controls.
- Protected response quality with a human-in-the-loop review gate, expanding automation only as measured accuracy justified it rather than exposing clients to unreviewed model output.
- Adapted a finance-focused LLaMA3 model with PEFT/LoRA adapters over a quantized base served locally through Ollama, synthesizing technical, fundamental, and sentiment signals into automated research reports.

PROJECTS

Generative AI Engineering on Databricks [Databricks, Vector Search, Unity Catalog, LLM-as-judge]
- Built a RAG application on Databricks Vector Search over Unity Catalog-governed data, orchestrating multiple models in one workflow and adding an agent-as-evaluator (LLM-as-judge) loop to score answer quality before release.

AI Agent Skills and Tooling for Claude [Claude, agent skills, tool orchestration, Python]
- Designed custom Claude agent skills that deliver targeted business requests end to end: a job-search agent that orchestrates multiple job-board and data connectors with scoring rules and cost guardrails, and a resume agent that compiles and verifies its own output before delivery.

Self-Hosted AI Job-Application Pipeline [GCP Compute Engine, Docker Compose, Caddy, Vertex AI, Supabase, n8n]
- Eliminated a daily manual workflow by architecting a 10-stage automation pipeline that ingests postings from four job sources, tailors a resume to each through a Vertex AI Gemini agent, renders LaTeX to PDF, verifies the compiled page count, and emails a ranked digest unattended every morning.
- Separated machine and user trust models by authenticating inference through a least-privilege service account while delegating Drive, Docs, and Gmail access through OAuth 2.0 user consent, and resolved the integration by isolating HTTP status semantics (401 to API enablement, 404 to a regional endpoint mismatch) rather than by trial and error.

Small Business Index Analysis and Forecasting [PySpark, Apache Spark, ETL, XGBoost, Streamlit]
- Produced forecast-ready datasets from 300,000+ records of raw sales data spanning 5 years by building a PySpark ETL pipeline that imputes missing values, detects outliers via interquartile range, applies Min-Max normalization, and one-hot encodes categorical features for distributed processing.
- Explained 90% of variance (R-squared ≈ 0.90) forecasting the Fiserv Small Business Index across 20 NAICS sectors and 50 U.S. states by benchmarking six ML models (XGBoost winner).

Airbnb Price Recommendation Model [MLOps, SQLite3, Docker, DagsHub, Streamlit]
- Predicted Airbnb listing prices at 92% of variance (R-squared ≈ 0.92) by processing raw data with SQLite3, engineering features from location, property type, and amenity attributes, and hyperparameter-tuning regression models.

NYC Taxi Tipping Pattern Analysis [Python, Pandas, Feature Engineering, Scikit-learn]
- Uncovered the drivers of passenger tipping behavior by processing 3 million+ NYC taxi trip records in Pandas, joining external borough-income and weather datasets and engineering features across passenger, trip, and socioeconomic dimensions.

CERTIFICATIONS

- Generative AI Engineering with Databricks, Databricks Academy (course completed) — 2026
- AWS Certified Cloud Practitioner (EC2, S3, Lambda, ECS, EKS, DynamoDB, SQS, IAM, VPC) — Aug 2026
- Monte Carlo 101, Certificate of Completion, Monte Carlo (data and AI observability) — Oct 2026

EDUCATION

M.S. in Engineering Science (Data Science), SUNY at Buffalo, NY — Jan 2024 – May 2025
B.E. in Computer Science (AI & ML), Annamalai University, India — Jul 2019 – May 2023

SKILLS

Programming Languages: Python, SQL, JavaScript, PySpark, R, C/C++, MATLAB, Haskell

Software Engineering: REST API design (FastAPI, Flask), React frontends, Git, GitHub Actions, CI/CD, pytest, vitest, unit/integration/regression testing, automated code generation, Agile delivery, Jira

Data Engineering: ETL/ELT, batch and event-driven pipelines, star-schema dimensional modeling, 3NF normalization, query optimization, idempotent loads and deduplication, data validation and regression testing, medallion architecture (bronze/silver), audit logging and load reconciliation, cron scheduling, DAG workflow orchestration (Apache Airflow, Google Cloud Composer, n8n), dbt

Databases & Storage: PostgreSQL, MySQL, SQLite3, Supabase, Snowflake (Dynamic Tables), ChromaDB, Neo4j

Data & AI Observability: Monte Carlo (freshness, volume, schema, and field-quality monitors), OpenTelemetry tracing, Arize Phoenix, LLM retrieval evaluation (hit rate@k, refusal accuracy)

Cloud & Infrastructure: Google Cloud Platform (Cloud Storage, Cloud Composer, Compute Engine, Vertex AI, IAM), Microsoft Azure, AWS, BigQuery, Apache Spark, Databricks (Vector Search, Unity Catalog), Docker, Docker Compose, Kubernetes, Terraform, Linux, reverse proxy and TLS (Caddy), OAuth 2.0, VPS self-hosting

Machine Learning: Scikit-learn, XGBoost, PyTorch, TensorFlow, Keras, Pandas, NumPy, feature engineering, model benchmarking and hyperparameter tuning, experiment tracking (DagsHub), OpenCV

AI/LLM Engineering: Hugging Face Transformers, BERT, T5, LLaMA3, Gemini, PEFT/LoRA fine-tuning, quantization, RAG, embeddings, vector search, hybrid dense/BM25 retrieval, reciprocal rank fusion, Model Context Protocol (MCP), LLM-as-judge evaluation, Claude, nomic-embed-text, Qwen, LangChain, LangGraph, Ollama